Example 4.49 Prove that ${}^{10}C_2 + 2 \times {}^{10}C_3 + {}^{10}C_4 = {}^{12}C_4$

Solution:

$$\begin{aligned} {}^{10}C_2 + 2 \times {}^{10}C_3 + {}^{10}C_4 &= {}^{10}C_2 + ({}^{10}C_3 + {}^{10}C_3) + {}^{10}C_4 \\ &= ({}^{10}C_2 + {}^{10}C_3) + ({}^{10}C_3 + {}^{10}C_4) \\ &= {}^{11}C_3 + {}^{11}C_4 \\ &= {}^{12}C_4 \end{aligned}$$

Example 4.50 If ${}^{(n+2)}C_7 : {}^{(n-1)}P_4 = 13 : 24$ find n.

Solution:

$$\begin{aligned} {}^{(n+2)}C_7 : {}^{(n-1)}P_4 &= 13 : 24. \\ \frac{{}^{(n+2)}C_7}{{}^{(n-1)}P_4} &= \frac{13}{24}. \\ \frac{(n+2)!}{(n-5)! \times 7!} \times \frac{(n-5)!}{(n-1)!} &= \frac{13}{24}. \\ \frac{(n+2)(n+1)n(n-1)!}{(n-1)! \times 7!} &= \frac{13}{24}. \\ (n+2)(n+1)(n) &= \frac{13}{24} \times 7! = \frac{13}{24} \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ (n+2)(n+1)(n) &= 13 \times 14 \times 15 \\ n+2 &= 15 \Longrightarrow n = 13. \end{aligned}$$

Example 4.51 A salad at a certain restaurant consists of 4 of the following fruits: apple, banana, guava, pomegranate, grapes, papaya and pineapple. Find the total possible number of fruit salads.

Solution:
There are seven fruits and we have to select four fruits for the fruit salad. Hence, the total number of possible ways of making a fruit salad is ${}^7C_4 = {}^7C_3 = 35$.

Example 4.52 A Mathematics club has 15 members. In that 8 are girls. 6 of the members are to be selected for a competition and half of them should be girls. How many ways of these selections are possible?

Solution:
There are 8 girls and 7 boys in the mathematics club. The number of ways of selecting 6 members in that half of them girls (3 girls and 3 others) is ${}^8C_3 \times {}^7C_3 = 56 \times 35 = 1960$.

Example 4.53 In rating 20 brands of cars, a car magazine picks a first, second, third, fourth and fifth best brand and then 7 more as acceptable. In how many ways can it be done?

Solution:
The picking of 5 brands for a first, second, third, fourth and fifth best brand from 20 brands in ${}^{20}P_5$ ways. From the remaining 15 we need to select 7 acceptable in ${}^{15}C_7$ ways. By the rule of product it can be done in ${}^{20}P_5 \times {}^{15}C_7$ ways.